

Multi-Source Candidate Data Transformer: Technical Design

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1. Pipeline Breakdown (The DAG Workflow)

The system operates as a Directed Acyclic Graph (DAG), ensuring dependencies are processed systematically, preventing the field-scrambling seen in naive linear parsers. Each stage streams per-candidate, keeping cost linear to thousands of candidates.

- **Ingestion and Detect:** routing based on source signature (structured ATS/CSV vs. unstructured PDF/Notes).
- **Pre-Flight and Extract:** coordinate-aware segmentation isolates spatial regions in resumes, preventing column-mixing, followed by hybrid NER (GLiNER for prose, FlashText for canonical skill-trie mapping).
- **Normalize:** deterministic, context-free transformation to canonical formats (E.164, ISO-8601).
- **Identity Resolve:** NetworkX graph clustering using high-entropy keys.
- **Merge (Reduction Phase):** application of the authority matrix for conflict resolution.
- **Project and Validate:** decoupled projection layer with Pydantic-based schema guards.

2. Canonical Schema and Normalized Formats

Implementation covers ATS JSON (structured) and Resume PDF (unstructured). The canonical record below refines the default schema (`headline`, `links`, `years_experience`, `education` are derived/optional, keeping the core record lean).

- **Identity:** `candidate_id` (deterministic hash), `full_name` (Unicode NFKC normalized).
- **Contact:** `phones` (libphonenumber, E.164), `emails` (normalized string).
- **Temporal / Locational:** `experience` (dates in YYYY-MM), `location` (ISO-3166 alpha-2).
- **Taxonomy:** `skills` array as `[{name, confidence, sources[]}]`, mapped via RapidFuzz to an alias table.
- **Provenance:** an audit trail logging (field, source, method) for every transformation.

3. Merge Policy, Conflict Resolution and Confidence

- **Match keys:** identity resolution uses strictly exact-match keys (email, phone). Name keys are corroborating signals only, never edge-triggers, to prevent false merges.
- **Authority matrix:** conflicts are resolved via a fixed source-trust hierarchy (ATS > CSV > Resume > Notes). Losers are retained in the `provenance_history` buffer, never discarded.

$$Value_{final} = \arg \max_{v \in V} \left(\sum_{s \in S} T_s \cdot I(v, s) \right)$$

Confidence scoring (C_a): $C_a = \sum (w_i \cdot m_i \cdot \phi_i)$. Confidence aggregates source trust (w_i), method extraction fidelity (m_i), and cross-source consensus (ϕ_i).

4. Runtime Custom-Output Config

- **Pure projection:** the engine separates the canonical record from the output format. Configuration Ω (field selection, path remapping, per-field normalization, `on_missing`: null/omit/error) acts as an instruction set: $P = C \times \Omega$.
- **Validation:** before delivery, the output is passed through a Pydantic schema guardrail to ensure strictly deterministic interface compliance.

5. Edge Cases and Resilience

Also considered: duplicate/stale-record conflicts across sources, partial or missing-field records, and schema drift from new upstream fields.

- **Non-linear layouts:** handled by spatial coordinate-aware segmentation, ensuring field relationships are preserved.
- **Identity precision:** false merges are prevented by disabling fuzzy identity matching; only strict high-entropy key inter-sections trigger a cluster merge.
- **Multilingual variants:** Unicode NFKC plus transliteration yields stable match keys while preserving original display strings.
- **Atomic failure model:** malformed sources (truncated JSON, empty CSV) are handled via independent failure logging; only the failed branch terminates, leaving the rest of the DAG intact.

Deliberately Left Out (Under Time Pressure)

- **Probabilistic / fuzzy matching:** excluded to ensure 100% deterministic reproducibility.
- **Full-scale OCR:** limited to a break-glass fallback for unreadable text layers only.
- **Horizontal scaling:** architecture is optimized for high-performance single-machine batch processing to avoid distributed state complexities.